

3(a)	any three from: active site with a specific shape as part of a lysozyme molecule drawn and active site labelled or included in an annotation ; complementary shaped peptidoglycan drawn <u>and</u>, peptidoglycan / substrate, labelled or included in an annotation; enzyme-substrate complex drawn <u>and</u> labelled ; lysozyme with same shaped active site and, two products / NAM and NAG drawn (formed from hydrolysed peptidoglycan) ; AVP ; e.g. water shown enzyme active site labelled 'lock' and substrate labelled 'key' <i>max 2 if no ref. to specific example e.g. label for lysozyme or peptidoglycan or murein or the products NAG and NAM</i> <i>max 1 no labels but otherwise clearly lock and key or by gaining the enzyme-substrate complex mark if induced fit drawn</i>	3
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Question	Answer	Marks
3(b)	<i>allow murein for peptidoglycan</i> <i>must attempt similarities and differences to gain max</i> any three from: <i>similarities, both</i> 1 cause cell, lysis / bursting ; I ref.to destroying / killing, bacteria (both) 2 detail ; e.g. weaken / destroy / AW, cell wall act on peptidoglycan cell wall cell wall cannot withstand turgor pressure / AW <i>differences</i> <i>(lysozyme is, a protein / an enzyme / a catalyst, but)</i> 3 penicillin is, an antibiotic / an enzyme inhibitor / a beta-lactam ; <i>(lysozyme hydrolyses, glycosidic bonds / peptidoglycan molecules, but)</i> 4 penicillin, inhibits transpeptidase(s) ; A peptidase(s) 5 penicillin, prevents formation of / AW, peptide / cross, bridges / linkages ; R penicillin breaks cross bridges A ref. to autolysins e.g. 'cannot repair holes made by autolysins' 6 penicillin only effective when cell is, growing / synthesising cell walls, but lysozyme effective, at all stages / AW; 7 AVP ; e.g. lysozyme (much) larger molecule <i>ref. to lysozyme denatured v penicillin less affected by heat</i> <i>idea of lysozyme more affected by factors affecting enzyme</i> A examples lysozyme globular v penicillin contains (beta-lactam / 4-member) ring	3